

DC motor control by ultrasonic sensor

Program:

```
// defines pins numbers ultrasonic sensor.

const int trigPin = 9;

const int echoPin = 10;

// defines pins numbers motors.

const int motor = 8;

// defines variables

long duration;

int distance;

void setup() {

  pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output

  pinMode(echoPin, INPUT); // Sets the echoPin as an Input

  pinMode(motor, OUTPUT); // Sets the Motor as an Output

  Serial.begin(9600); // Starts the serial communication

}

void loop() {

  // Clears the trigPin

  digitalWrite(trigPin, LOW);

  delayMicroseconds(2);

  // Sets the trigPin on HIGH state for 10 micro seconds

  digitalWrite(trigPin, HIGH);

  delayMicroseconds(10);

  digitalWrite(trigPin, LOW);

  // Reads the echoPin, returns the sound wave travel time in microseconds

  duration = pulseIn(echoPin, HIGH);

  // Calculating the distance
```

```
distance= duration*0.034/2;  
// Prints the distance on the Serial Monitor  
Serial.print("Distance: ");  
Serial.println(distance);  
if(distance < 10){  
    digitalWrite(motor, HIGH);  
}else{  
    digitalWrite(motor, LOW);  
}  
}
```

Link for video:

<https://www.youtube.com/watch?v=1SGTYtD1e-k>

<https://www.youtube.com/watch?v=rbXG30tYBjY>

Wiring:

trig pin to digital pin 9

echo pin to digital pin 10

dc motor positive wire to digital pin 8